The documentation and process conversion measures necessary to comply with this revision shall be completed by 21 August 1993

INCH-POUND

MIL-S-19500/228G
21 May 1993
SUPERSEDING
MIL-S-19500/228F
19 April 1991

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, RECTIFIER,
TYPES 1N3611, 1N3612, 1N3613, 1N3614, 1N3957,
JAN AND JANTX

These devices are no longer active for new design, prefered devices are 1N5614, 1N5616, 1N5618, 1N5620, 1N5622 on MIL-S-19500/427

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for 1.0 ampere silicon rectifier diodes. Two levels of product assurance is provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1.

1.3 Maximum ratings.

Types	V_{RWM}	I_O	
		At: T_A = +100°C 1/ 2/	At: T_A = +150°C 1/ 2/
	V(pk)	A dc	mA dc
1N3611	200	1	300
1N3612	400	1	300
1N3613	600	1	300
1N3614	800	1	300
1N3957	1000	1	300

1/ From I_O rating is independent of heat sinking, special mounting, or leads of the device.
2/ Derate linearly at 13.3 mA°C betweem T_A = +100°C and T_A = +175°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Electronics Supply Center, DESC-ECT, 1507 Wilmington Pike, Dayton, OH 45444-5270 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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1.3 Maximum ratings. continued

Types	I_{FSM} T_A = +25°C I_O = 1 A dc t_p = 8.0 ms	Barometric pressure (reduced)	T_{STG} and T_J	$R_{\theta JL}$ L = 3/8 inch
	A(pk)	mmHg	°C	°C/W
1N3611	30	8	-65 to +175	38
1N3612	30	8	-65 to +175	38
1N3613	30	8	-65 to +175	38
1N3614	30	54	-65 to +175	38
1N3957	30	87	-65 to +175	38

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

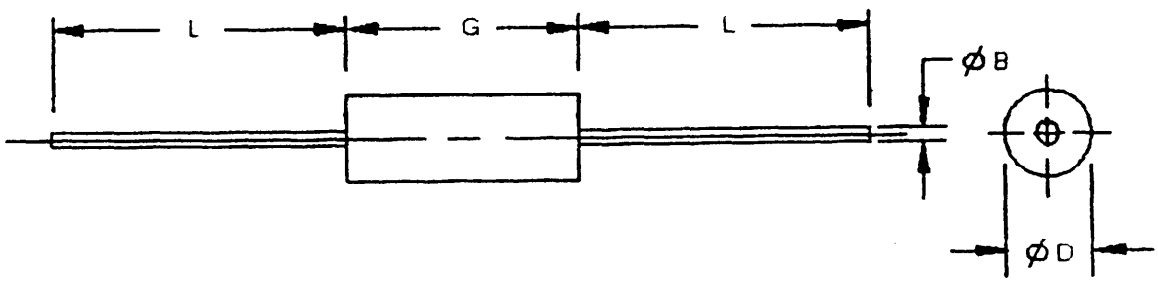
2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
ØB	0.025	0.034	0.64	0.86	3
ØD	0.060	0.110	1.52	2.79	
G	0.140	0.205	3.56	5.21	
L	0.600	1.500	15.24	38.10	4

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimension ØB shall be measured at the largest diameter.
4. Dimension G shall include the sections of the lead over which the diameter is uncontrolled. This uncontrolled area is defined as the zone between the edge of the diode body and extending .050 inch (1.27 mm) onto the leads.

FIGURE 1. Physical dimensions.

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3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and on figure 1 herein. No lead (Pb) shall be used in the construction of the die bonds.

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-S-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.3.2 Diode construction. These devices shall be constructed utilizing non-cavity double plug construction with high temperature bonding between both sides of the silicon die and terminal pins.

3.4 Marking. Devices shall be marked as specified in MIL-S-19500.

3.5 Polarity. The polarity shall be indicated with a contrasting color band to denote the cathode end.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.3 Screening (JANTX level only). Screening shall be in accordance with MIL-S-19500 (table IIb), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see TABLE II of MIL-S-19500)	Measurement
	JANTX level
9	Not applicable
11	I_{R1} and V_{F1}
12	See 4.3.2
13	Subgroup 2 of table I herein. ΔI_{R1} = 100 percent of initial reading or ±50 nA dc, whichever is greater. ΔV_{F1} = ±0.1 V dc.

4.3.2 Power burn-in conditions. Power burn-in conditions are as follows: Method 1038, condition B of MIL-STD-750, T_A = room ambient as defined in the general requirements of paragraph 4.5 of MIL-S-19500; V_{RWM} = full rated V_{RWM} (see 1.3); f = 50 to 60 Hz; I_O = 1 A dc; 96 hours.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JANTX, and JANTXV) of MIL-S-19500, and and as follows. Electrical measurements (endpoints) and delta requirements shall be in accordance with the applicable steps of table II herein.

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4.4.2.1 Group B inspection, table IVb of MIL-S-19500.

Subgroup	Method	Conditions
B3	1026	T_A = 150°C° ±5°C, I_O = 300 mA dc, V_R = rated V_{RWM} (see 1.3), t = 340 hours, f = 60 Hz.
B5	3101 or 4081	25°C ≤ T_L ≤ 35°C (recorded before test is performed), $R_{\theta JL}$ ≤ maximum rated $R_{\theta JL}$ (see 1.3).
B6	1032	T_A = +175°C, t = 340 hours.
B7	4066	I_{FSM} = 30 A(pk), 10 surges of 8.3 ms each at 1 minute intervals, I_O = 1 A dc, V_{RWM} = rated V_{RWM} (see 1.3), T_A = 25°C. LTPD = 10

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500 and as follows. Electrical measurements (endpoints) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.3.1 Group C inspection table V of MIL-S-19500.

Subgroup	Method	Conditions
C2	2036	Test condition A, weight = 4 pounds, t = 15 s.
C2	2036	Test condition E.
C5	1001	V_{RWM} = rated V_{RWM} (see 1.3), pressure = rated pressure (see 1.3), t = 1 minute, LTPD = 15.
C6	1026	T_A = 150°C ± 5°C, I_O = 300 mA dc, V_{RWM} = rated V_{RWM} (see 1.3), f = 60 Hz.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows.

4.5.1 Burn-in and steady-state operation life tests. These tests shall be conducted with a half-sine waveform of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conduction angle of the rectified current shall be neither greater than 180°, nor less than 150°.

4.5.2 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Forward voltage	4011	I_F = 1 A dc (pulsed) (see 4.5.2); duty cycle = 2 percent maximum	V_{F1}	0.6	1.1	V pk
Reverse current leakage	4016	V_R = rated V_{RWM} (see 1.3) DC method	I_{R1}		1.0	µA dc
Subgroup 3						
High temperature operation		T_A = 150°C				
Reverse current leskage	4016	V_R = rated V_{RWM} (see 1.3) DC method	I_{R2}		300	µA dc
Low temperature operation		T_A = -65°C				
Forward voltage	4011	I_F = 1.0 A dc (pulsed) (see 4.5.2)	V_{F2}	0.6	1.5	V dc

1/ For sampling plan, see MIL-S-19500.

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TABLE II. Groups A, B and C electrical measurements. 1/ 2/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Forward voltage	4011	I_F = 1.0 A dc pulse width = 8.3 ms maximum; duty cycle = 2 percent maximum;	V_{F1}	0.6	1.1	V dc
2.	Reverse current	4016	DC method; V_R = rated (see 1.3)	I_{R1}		1.0	µA dc
3.	Forward voltage	4011	I_F = 1.0 A dc pulse width = 8.3 ms maximum; duty cycle = 2 percent maximum	ΔV_{F1}	±0.1 V dc maximum		
4.	Reverse current	4016	DC method; V_R = rated (see 1.3)	ΔI_{R1}	±50 nA dc or 100 percent of initial value whichever is greater		

1/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1 and 2.

b. Subgroup 3, see table II herein, steps 1, 2, 3, and 4.

c. Subgroup 6, see table II herein, steps 1 and 2.

d. Subgroup 7, see table II herein, steps 1 and 2.

2/ The electrical measurements for table V of MIL-S-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1 and 2.

b. Subgroup 3, see table II herein, steps 1 and 2.

c. Subgroup 5, see table II herein, step 2.

d. Subgroup 6, see table II herein, steps 1 and 2.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as specified (see 3.3.1).

c. Type designation and product assurance level.

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI, AV
Air Force - 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Navy - EC

Agent:
DLA - ES

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INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-S-19500/228G	930521

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, DIODE, SILICON, RECTIFIER, TYPES 1N3611, 1N3612, 1N3613, 1N3614, 1N3957 JAN AND JANTX

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY	
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c. ADDRESS (Include Zip Code) DEFENSE ELECTRONIC SUPPLY CENTER, DESC- ECT 1507 WILMINGTON PIKE, DAYTON, OHIO 45444-5270	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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